BANASTHALI UNIVERSITY

Project title:-CrimeLink (Online Crime Reporting Platform)

BCA 6th Semester Group Project

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1.INTRODUCTION

The Online CrimeReporting System (CrimeLink) is a web-based application designed to facilitate the reporting and management of criminal activities. This system provides a platform for citizens to report crimes efficiently and securely while enabling law enforcement agencies to respond effectively.

1.1 Purpose

The purpose of this online crime reporting system is to simplify and expedite the crime reporting process. It aims to provide a platform for citizens to report crimes online with ease, upload evidence, and track the status of their reports in real-time.

The purpose of the Online Crime Management System is to modernize and improve the efficiency of police departments by transitioning from a manual record-keeping system to a computerized one.

1.2 Scope

CrimeLink serves citizens and police officers.This system is designed to be a comprehensive platform that transforms how crime reporting and management are conducted.The scope of this system encompasses several key areas aimed at automating and improving the processes within police departments and with citizens.Here are the main aspects of its scope:-

A. Citizens can report crimes with evidence and track the status of their reports.
B. Police officers can review cases, assign them to teams, and update statuses.

CrimeLink is designed as a small-scale project aimed at demonstrating the feasibility of an online crime reporting system. **It is primarily intended for academic purposes** and can be expanded for real-world implementation with additional features and security enhancements

2.Overall Description

2.1 Product Perspective

The Online Crime Reporting Platform is designed as a centralized, user-friendly, and secure web-based system to streamline the process of reporting, managing, and tracking crimes. It bridges the gap between citizens, law enforcement agencies, and administrative entities, fostering efficient communication and resolution of issues.

This platform modernizes traditional, often cumbersome methods of crime reporting—such as physical visits to police stations or phone calls—by providing a digital solution that is accessible, transparent, and convenient. The system reduces paperwork, minimizes response times, and enables citizens to report crimes, upload evidence, and track their cases in real-time.

The system forms a part of broader efforts to digitize and modernize law enforcement operations. It integrates seamlessly with existing police databases and government communication networks, while also supporting advanced features like crime categorization and analysis. The platform is accessible via both desktop and mobile devices, ensuring inclusivity and ease of use for a diverse range of users.

With features like real-time status tracking, AI-powered chatbot assistance, and an emergency SOS function, this system aims to enhance public safety, improve user satisfaction, and optimize the efficiency of law enforcement processes.

2.2 Product Function

The Online Crime Reporting Platform provides the following core functions:

2.2.1. User Registration and Authentication

A. Secure account creation for citizens, law enforcement personnel, and administrators.
B. Role-based access controls to ensure proper authorization and system integrity.

2.2.2. Crime Reporting

A. User-friendly forms for reporting various types of crimes, including theft, harassment, fraud, and more.
B. Option to report crimes anonymously, ensuring user privacy.
C. Support for uploading evidence such as photos, videos, and documents to strengthen case validity.

2.2.3. Case Management

A. Automated case ID assignment for every report filed.
B. Real-time tracking of case progress, with statuses such as "Under Review," "In Progress," and "Closed."

2.2.4. Communication Channels

A. AI-powered chatbot for answering frequently asked questions and assisting users in navigating the platform.

2.2.5. Analytics and Reporting

A. Tools for generating detailed statistical reports and identifying crime trends.

B. Dashboards for administrators and law enforcement to monitor and manage cases effectively.

2.2.6. Emergency Assistance

A. Quick access to an SOS feature for users in critical situations, notifying authorities with location details.
B. Immediate alerts sent to the nearest police station or designated emergency responders.

2.2.7. Data Security and Privacy

A. End-to-end encryption for all sensitive data, ensuring secure storage and transmission.
B. Compliance with legal and regulatory standards for data protection.
C. Regular audits and monitoring to prevent misuse, ensure transparency, and maintain system integrity.

2.3 User Characteristics

2.3.1.Citizen Users

Citizen users are individuals who utilize the **CrimeLink platform** to report crimes, track case progress, or seek emergency assistance. They come from diverse backgrounds and may include:

A. **General Public** – Citizens reporting crimes such as theft, harassment, fraud, or cybercrime.
B. **Victims of Crimes** – Individuals seeking justice who may choose to report incidents for safety reasons.
C. **Witnesses & Bystanders** – People who observe criminal activities and wish to report them.
D. **Concerned Community Members** – Individuals who want to contribute to public safety by sharing relevant information.

2.3.2.Police Officers

Police Officers play a critical role in handling reported crimes and ensuring justice. This group includes:

A. **Police Officers** – Review cases, verify crime reports, assign investigations, and update case statuses.
B. **Emergency Responders** – Handle high-priority SOS alerts and take immediate action in critical situations.

3. Non-Functional Requirements

3.1. Performance

A. The system must handle at least 500 concurrent users with minimal latency (<3 seconds response time)
B. Efficient database queries to support real-time updates.

3.2. Security

A. Encrypted data transmission using HTTPS
B. Secure authentication with MFA and password hashing.
C. Regular audits for vulnerability assessments.

3.3.Scalability

A. Designed to support increasing user load and data storage without degradation in performance.
B. Flexible architecture to integrate new features and third-party services.

3.4.Usability

A. Intuitive, role-specific interfaces for citizens, police, and admins.
B. Accessible on all major devices with responsive design.
C. Clear navigation and guided processes for seamless user experience.

4. Functional Requirements

4.1 User Registration & Authentication

Secure account creation with MFA, role-based access, and password recovery.

4.2 Crime Reporting

Guided forms with file uploads, geotagging, and acknowledgment via unique report ID.

4.3 Case Tracking

Check the current status and progress of the case.

4.4 Emergency SOS

Quick alerts and live location sharing

4.5 Admin Panel

Manage users, assign cases, view analytics, and monitor logs.

5. Requirement Specifications

5.1 User Interface

The platform ensures a user-friendly experience for citizens, police officers, and administrators.

5.1.1. Login & Registration

A. Intuitive forms with role-based login (Citizen, Police, Admin).
B. Fields: Username, Password, Email, Role Selection.

5.1.2. Dashboards

Citizen Dashboard:

A. **Report Crimes** – Submit crime reports with details and evidence.
B. **Track Case Status** – Monitor the progress of submitted reports.
C. **Interact with the Chatbot** – Get assistance for FAQs and crime categorization.

Police Dashboard:

A. **View Assigned Cases** – Access and review crime reports allocated to them.
B. **Update Case Statuses** – Modify report progress (e.g., Under Review, In Progress, Closed).
C. **Manage Users and Cases** – Oversee citizen, police, and admin accounts.
D. **Access Analytics and System Logs** – Monitor system activity, generate reports, and track performance.

5.1.3.Crime Reporting:

Comprehensive form with:

A. **Crime Type**: Dropdown menu (e.g., theft, harassment).
B. **Date/Tim**e: Input fields.
C. **Evidence Upload**: Options for images, videos, or documents.

5.1.4.Additional Features:

A. **Case Tracking**:Status updates (e.g., "Under Review," "Closed")

B. **Chatbot**:Guides reporting and answers FAQs.

5.2. Hardware Interface

5.2.1. Client-Side Requirements

A. **Supported Devices**: Desktop Computers, Laptops, Smartphones, andTablets.
B. **Minimum Specifications**:Processor: Dual-core (Intel i3 or equivalent) or higher.
C. **RAM**: Minimum 2GB.
D. **Storage**: At least 500MB of free space for caching browser data.
E. **Display Resolution**: 1024x768 pixels or higher.
F. **Internet Speed**: Minimum 2 Mbps for a smooth user experience.
G. **Browser Compatibility**: Chrome, Firefox, Edge, Safari (latest versions).

5.2.2. Server-Side Requirements

A. **Processo**r: Quad-core (Intel Xeon or AMD Ryzen) or higher.
B. **RAM**: Minimum 8GB (for handling concurrent users and processing large amounts of data).
C. **Storage**: At least 100GB SSD (to store reports, evidence files, and logs).
D. **Bandwidth**: Minimum 100 Mbps (to support multiple users simultaneously).
E. **Database Suppor**t: MySQL or PostgreSQL with optimized indexing for fast queries.

5.2.3 Additional Peripherals (For Evidence Submission)

To facilitate crime reporting with supporting evidence, the following peripherals are recommended:

A. **Scanners or Cameras** – For uploading clear scanned documents and photographic evidence.
B. **Smartphones with GPS Support** – Enables geotagging of crime reports for precise location tracking.

5.3. Software Interface

5.3.1. Backend

PHP for server-side scripting.

5.3.2. Frontend

HTML, CSS, JavaScript for interactivity and UI design.

5.3.3. Database

MySQL with tables for:

A. **Users**: UserID, Role, Email, Password.
B. **Cases**: CaseID, CrimeType, Status, EvidencePath.
C. **Notifications**: Email/SMS notifications.

5.4. Communication Interface

5.4.1. Secure Interaction

HTTPS protocols and encrypted form submissions.

5.4.2 Notifications

A. **Email**: SMTP for sending case updates.
B. **SMS**: Twilio or equivalent API.

5.4.3 Chatbot Communication

A. **Python API**-driven text-based responses.

6.Introduction

The **Software Design Specification (SDS)** document serves as a blueprint for the technical implementation of a software system. It outlines the architectural framework, design methodologies, and technical specifications needed to build, test, and deploy the system effectively.

6.1Purpose

The purpose of this document is to provide a detailed design for the Online Crime Reporting System. This document outlines the architectural framework, design patterns, and workflows required for implementing the system effectively.This document also includes details of the database, user roles, chatbot integration, and emergency features.

6.2Scope

The Online Crime Reporting System provides a web-based solution for citizens to report crimes, upload evidence, and track the status of their cases. The system caters to three primary roles:

1. **Citizens**: Can report crimes, upload evidence, remain anonymous if needed, and track case updates.
2. **Police Officers**: Review cases, update statuses, and manage evidence.

7.1 User Case Diagram:

7.1.1 Citizen Interaction

A. **Actor**: Citizen (User)
B. **System**: Online Crime Reporting Application
   1. Register/Login – Users create accounts and log in (*Validates user credentials*).
   2. Report a Crime – Citizens submit crime details (*Assigns unique case ID*).
   3. Upload Evidence – Allows uploading images, videos, or documents.
   4. Track Case Status – Users monitor progress (*Under Review, In Progress, Closed*).
   5. Administer User Account – Manage profile and security settings.
   6. Report Emergency SOS – Sends an immediate distress alert.

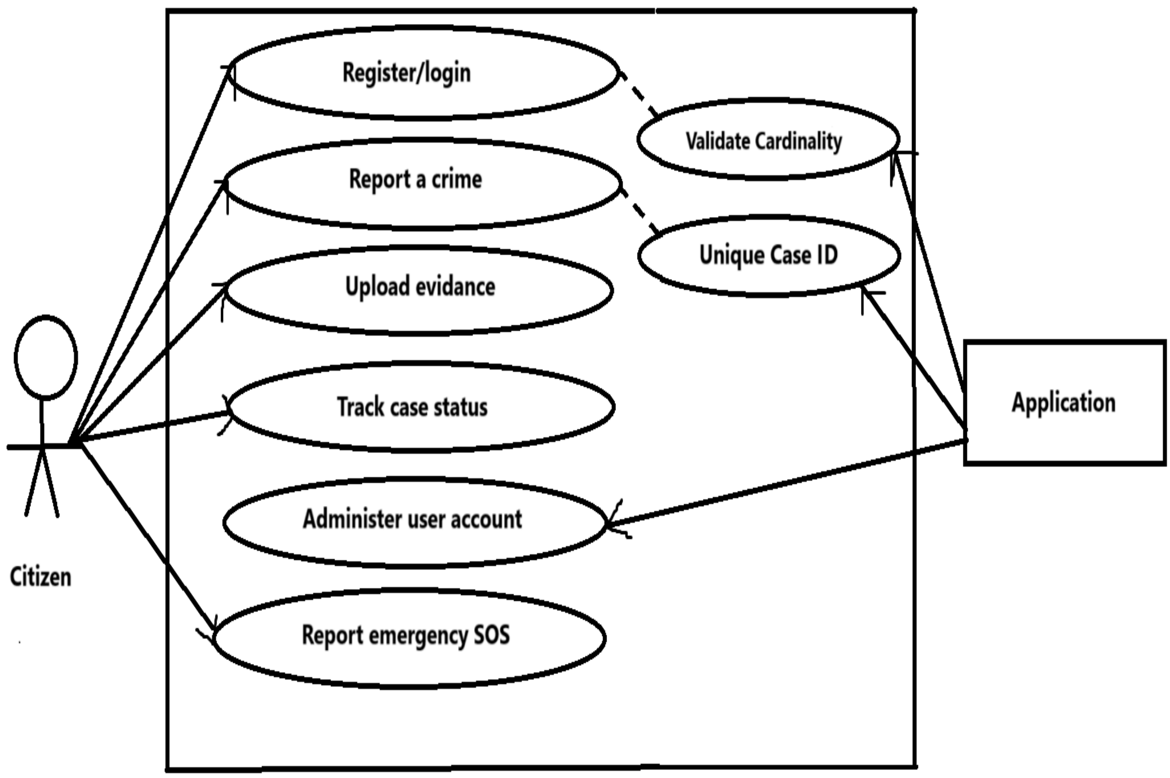

Fig 1: Interaction between citizen and application

7.1.2 Police Interaction

A. **Actor:** Police (Authorized Officer)
B. **System**: Online Crime Reporting Application
   1. Review Crime Reports – View and analyze reported cases.
   2. Allocate Cases – Assign cases to appropriate officers for investigation.
   3. Update Case Status – Modify case progress (*Under Review, In Progress, Resolved*).
   4. Delete Cases – Remove duplicate or invalid reports.
   5. Close Cases – Mark cases as resolved and archive them.

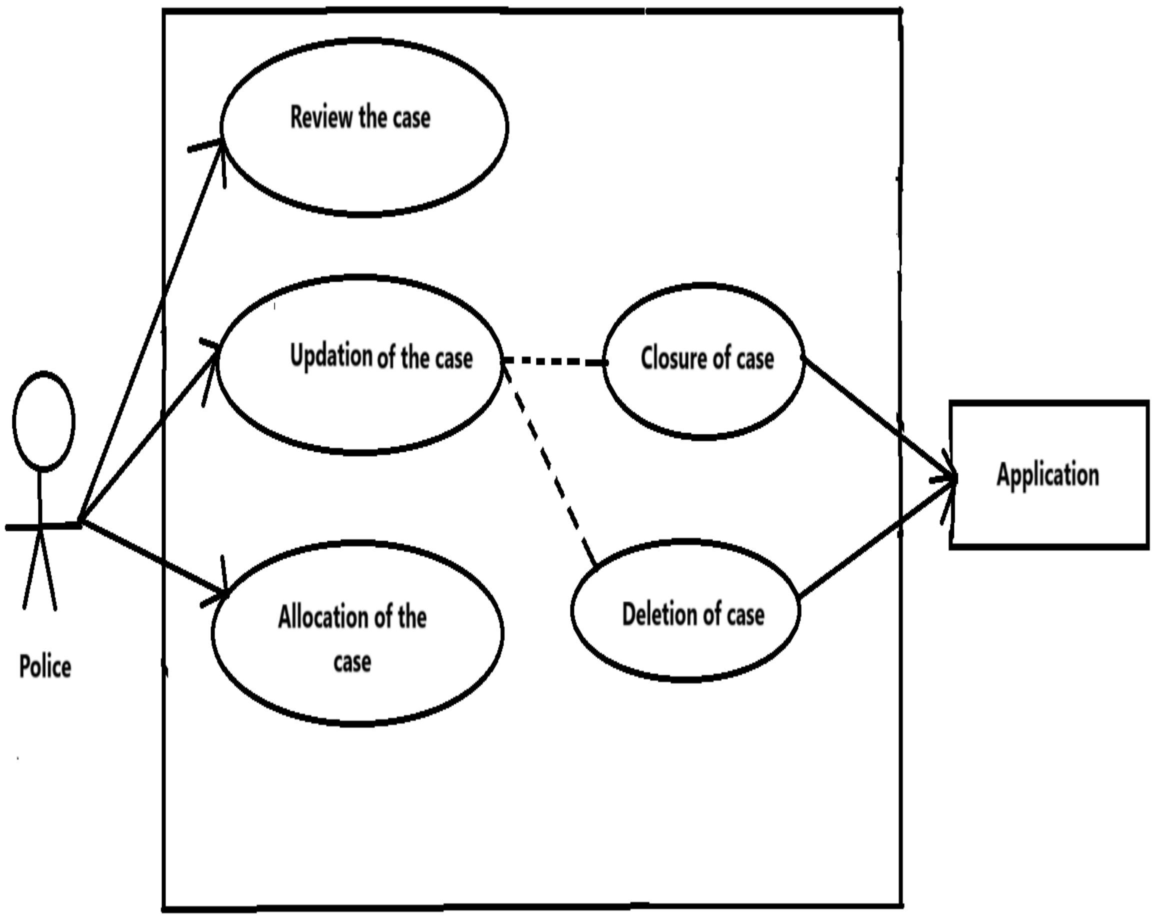

Fig 2: Interaction between police and application

7.2ER Diagram

The **ER Diagram** for the **Online Crime Reporting System** visually represents how different entities interact with each other in the database. It defines the relationships among users (citizens, police, and admins), crime reports, evidence and status tracking.It ensures that all components work together seamlessly, making crime reporting **efficient, secure, and transparent**.

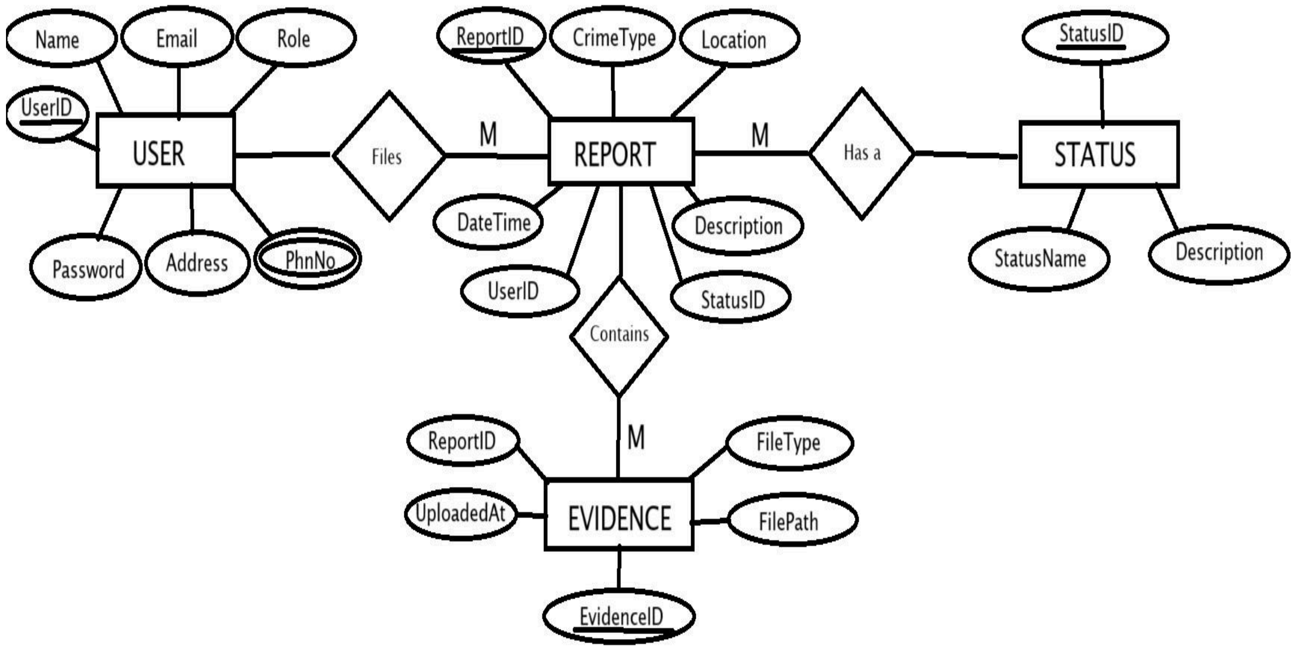

Fig : ER Diagram of the crime reporting system

7.2.1 Table Representation of ER Diagram

User Table

UserID	Name	Email	Password	Role	PhnNo	Address

Report Table

ReportID	CrimeType	Location	DateTime	Description	UserID	StatusID

Status Table

StatusID	StatusName	Description

Evidence Table

EvidenceID	ReportID	FilePath	FileType	UploadedAt

7.3 Activity Diagram Citizen

7.3.1 Citizen

A. **User Logs In** – Citizens, police, and admins authenticate and access their dashboards.
B. **Crime Reporting** – Citizens submit crime details, upload evidence, and receive a Case ID.
C. **Admin Review** – Admin verifies the report and assigns it to a police officer.
D. **Police Investigation** – Officer updates case status and investigates the report.
E. **Citizen Tracks Status** – Users check updates and provide additional evidence if needed.

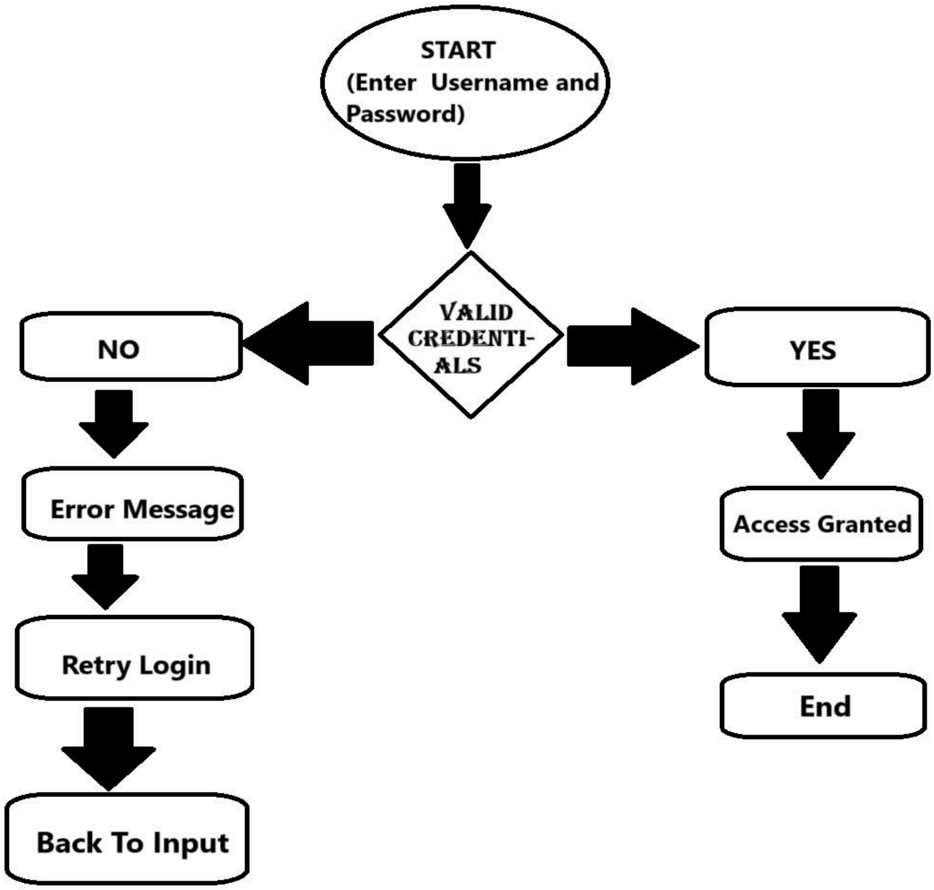

Fig : Activity diagram of the Crime reporting system

7.3 Activity Diagram

7.3.2 Police Officer

1. **Login –** The police officer logs into the system to access assigned cases.
2. **Conduct Investigation –** The officer reviews reports, gathers evidence, and interviews witnesses.
3. **Update Case Status –** The officer marks the case as *In Progress, Awaiting Evidence,* or *Closed.*
4. **Close Case –** If the case is resolved, it is officially closed.
5. **Notify Citizen & End –** The system informs the complainant, and the process is completed.

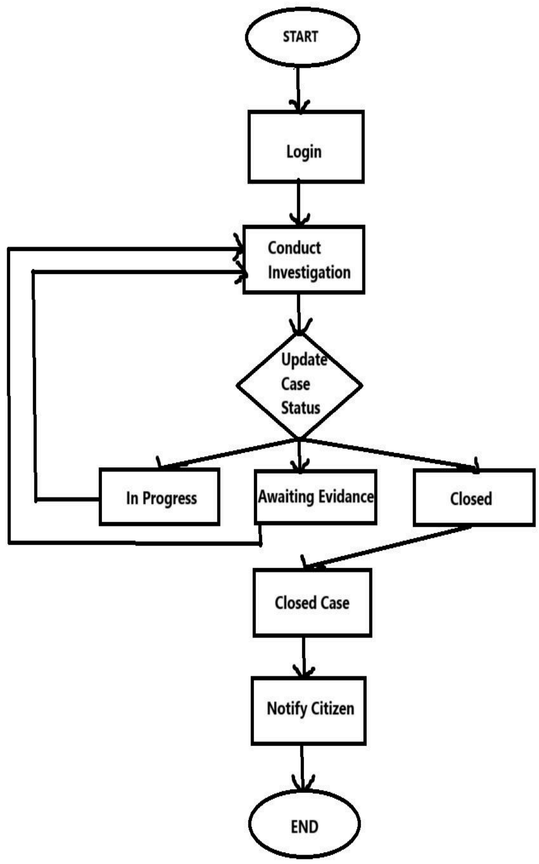

Fig : Activity diagram of Police

7.5Sequence diagram

7.5.1. Citizen

The diagram illustrates the sequence of interactions between the Citizen, Device, and Database while reviewing, allocating, updating, deleting, or closing a case.

A. **Register/Login** – Citizen registers or logs in to the system.
B. **Report a Crime** – Fills out crime details (type, location, time, evidence).
C. **Upload Evidence** – Submits supporting documents (images, videos, files).
D. **Track Case Status** – Checks case progress and status updates.
E. **Receive Notifications** – Gets alerts via email/SMS for updates.
F. **Report Emergency SOS** – Triggers an SOS alert for immediate police action.

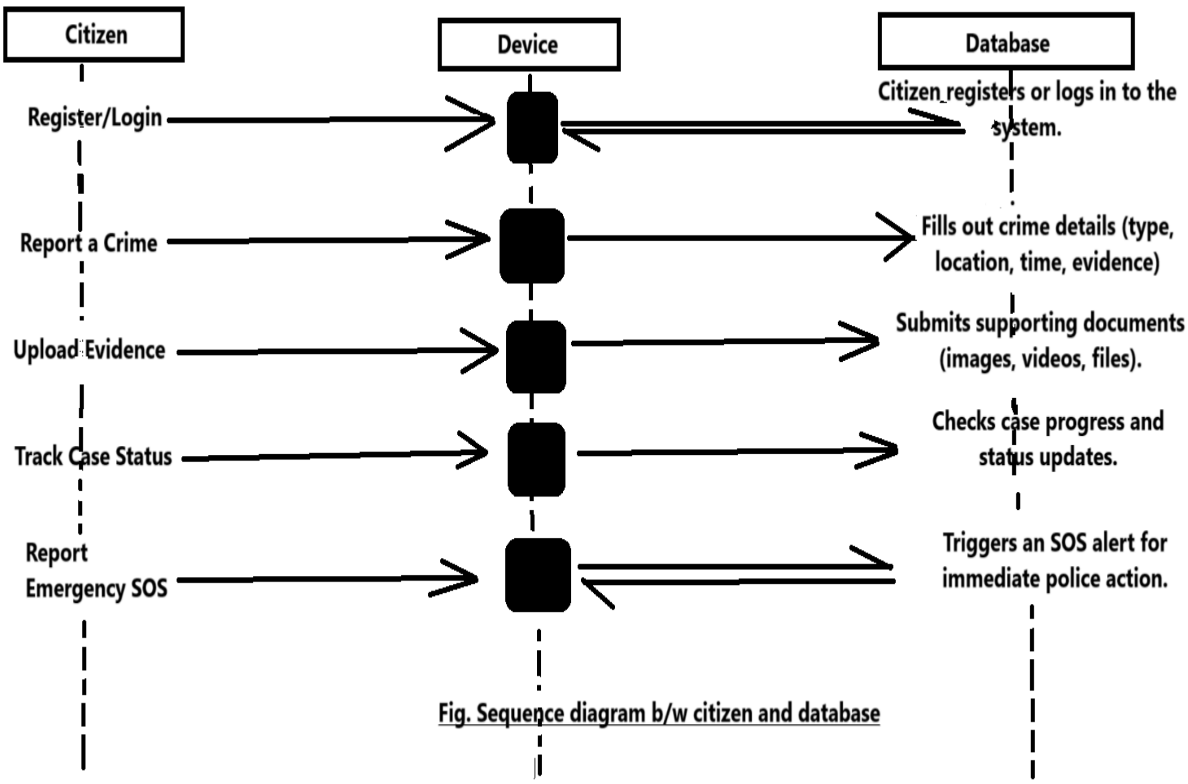

Fig. Sequence diagram b/w citizen and database

7.5.2 Police Officer

The diagram illustrates the sequence of interactions between the Police Officer, System, and Database while reviewing, allocating, updating, deleting, or closing a case.

A. **Police logs into the system –** Authentication is verified.
B. **Police reviews crime reports –** Retrieves reports from the database.
C. **Police allocate a case –** Assigns the case to an officer.
D. **Police updates case status –** Modifies the case progress.
E. **Police deletes a case (if invalid) –** Removes duplicate or fake reports.
F. **Police closes the case –** Marks it as resolved and archives it.

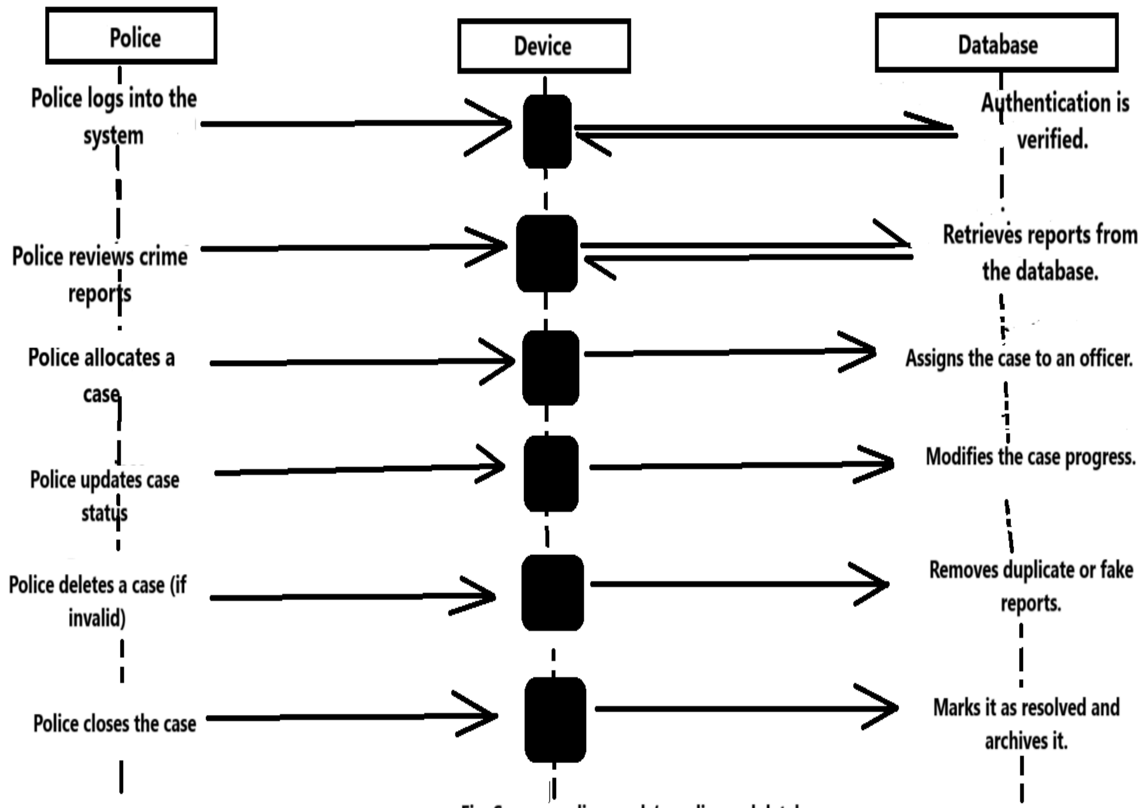

Fig. Sequence diagram b/w police and database

7.5Database Description

The CrimeLink database is structured to efficiently store, manage, and retrieve information related to crime reports, users, evidence, and case statuses. The database is built using MySQL and follows a relational database model, ensuring secure and organized data storage.

7.5.1 Key Database Tables & Their Purpose:

A. Users Table:-
- Stores details of all users (citizens, police officers, and administrators).
- Ensures role-based access for different users.

B. Reports Table
- Stores crime reports submitted by citizens.
- Each report is assigned a unique **Report ID** and linked to a user.

C.Evidence Table

- Stores uploaded evidence such as images, videos, or documents.
- Each piece of evidence is linked to a specific crime report.

D.Status Table

- Tracks the progress of each crime report (**Under Review, In Progress, Closed**).
- Ensures real-time updates for users.

7.5.2.Database Relationships (One-to-Many)

A. One user can submit multiple reports.
B. One report can have multiple evidence files.
C. One report is linked to one status at a time.

The database follows **data integrity principles** by using **Primary Keys (PK) and Foreign Keys (FK)** to maintain proper relationships and avoid redundancy.

7.6Conclusion

Key benefits of this system include:

A. **Faster and** The **CrimeLink Online Crime Reporting System** is designed to provide a **secure, efficient, and user-friendly platform** for reporting and managing criminal activities. By leveraging technology, the system **simplifies the crime reporting process** for citizens while enabling law enforcement agencies to **respond more effectively**.
B. With features like **real-time status tracking, evidence submission, and role-based access**, this system **eliminates delays and enhances transparency** in crime investigation processes. Citizens can **submit reports quickly**, **track their case progress**, and receive timely **notifications** without visiting a police station.
C. For law enforcement agencies, CrimeLink improves **case management and resource allocation**. Officers can **review reports, assign cases, update statuses**, and **respond to emergencies efficiently**. Administrators can oversee **user management, system performance, and generate analytical reports** for better decision-making.
D. **more efficient crime reporting** without the need to visit police stations.
E. **Real-time status tracking** to keep users informed about their case progress.
F. **Secure storage of data and evidence** to prevent manipulation.
G. **User-friendly interface** for citizens, police officers, and administrators.
H. **Scalability and future enhancements** such as AI crime prediction and mobile app integration.

By implementing this system, **law enforcement agencies can improve response times, reduce paperwork, and enhance overall public security**.

This documentation serves as a comprehensive guide for understanding the **technical and functional aspects** of the CrimeLink system, ensuring a **well-structured and efficient approach** to online crime reporting.